

Meeting Transcript (Engineering Sync – 2024-01-31):

Alice: “Today we'll start with status—how is the PDF ingestion pipeline progressing?”

Bob: “Almost done! We now tokenize PDFs using layout-aware parsers, so multi-column and table data is preserved.”

Eve: “I added tests with diverse PDFs—financial, academic, and contracts. The Go indexer handled 24k docs in ~50s.”

Tom: “LLM integration is next. We want agents that can route queries between Go and a fallback Python engine.”

Sara: “Frontend team finished the TypeScript demo. It now auto-detects doc types and shows latency per query.”

Alex: “I improved search precision for 'document type:pdf' and 'entity:contract'. A/B test results look great.”

Ada: “Reliability update: tested data loss scenarios and segment recovery—100% pass!”

Bob: “Bug: PDF with 12MB scanned tables triggers OOM. Root cause: we accidentally bypassed chunked upload.”

Santiago: “Let's prioritize that bug. Next sprint focus: semantic search, metrics, smarter query normalization.”

Tom: “Can we benchmark new Go routines? LLM agent orchestration is burning CPU on AE5 nodes.”

Lisa: “We found duplicate dict segments after rapid ingest. Will draft a cleanup PR.”

Jon (Eng. Mgr): Good morning! Let's call today's sprint retro to order. Thanks for joining on time.

Priya (Frontend): Morning, Jon! Morning all.

Alex (Backend): Hey everyone.

Rachel (QA Lead): Morning!

Jon: Let's kick off with sprint successes—Alex, can you start?

Alex: Sure! Search endpoint latency dropped by 15% after the Go profiler tuning. I want to thank Priya for flagging the bottleneck in the dashboard logs.

Priya: That was teamwork! Also, frontend deployed the new TypeScript UI for search filters. We resolved the weird Firefox dropdown bug.

Rachel: On QA, the PDF extraction edge cases you added, Alex, helped us catch two more parsing issues.

Jon: Awesome. What about CI/CD? Any blockers?

Rachel: Build pipeline had a false positive on the new SAST rules last Tuesday, but we traced it to outdated YAML.

Jon: Thanks, Rachel. Next—roll call of open issues. Let's try to keep answers brief.

Alex: We're tracking a rare crash when previewing multi-column PDFs in staging. Stack trace shows a panicking goroutine at line 416 in pdf_parser.go. Started after the async refactor.

Priya: I got a bug report from a customer saying that sometimes the new “export to CSV” button becomes unresponsive if they run more than 5 queries in a session.

Rachel: There's some flakiness in End-to-End tests for the annotation editor on Safari. Failure seems like a UI event timing issue.

Jon: Priya, can you get a screen recording and post it on the ticket?

Priya: Will do today.

Jon: Alex, action item: can you pair with Rachel this afternoon to triage the PDF preview crash?

Alex: Booked 2–3pm for that already!

Rachel: Same on my calendar, thanks.

Priya: I can patch the export button issue and add a debounce; will open a PR.

Jon: Thank you. Now, larger topics—our quarterly OKRs. Business wants us to increase time-to-data for first-time users by 20%. Any brainstorms on onboarding?

Rachel: Maybe we add a sample dataset wizard post-signup, preload a quick-start PDF and a test query so users see results in 15 seconds.

Priya: Love it. I'll mock up a stepper with four screens: upload > preview > search > download.

Alex: Backend can expose a /sample endpoint; can proxy to controlled S3 buckets.

Jon: Can one of you volunteer to drive this from design to deploy?

Priya: I can lead, at least through the prototype.

Alex: Happy to be co-pilot; ping me for API!

Jon: Thanks, both. Rachel, can QA get signoff before we push to prod?

Rachel: Yes, but we'll want staging users. Suggest 3–4 users outside this team to dogfood.

Jon: Love it. Now, quick check: Did anyone go to last Friday's "Tech Deep Dive" all-hands?

Priya: The CTO demonstrated a voice-search integration demo using Whisper. Pretty slick.

Alex: Anyone know which branch has the prototype?

Rachel: It's feature/whisper-integration, will merge back to dev after next QA sprint.

Jon: Anyone blocked or confused by shifting priorities, say, with v2 roadmap?

Priya: I'm still hazy on when the new PDF annotation system should be feature complete.

Jon: Good call. Updated roadmap hit Slack this morning, but I'll walk us through now: PDF annotation MVP slips from March to April for bandwidth.

Alex: That helps. We can focus on PDF cleanup tooling first, then UI.

Priya: I'll coordinate with design.

Jon: Anything else before we move to upcoming leave/OOO?

Rachel: I'm off next Friday for vacation.

Alex: I'll cover QA reviews for Rachel.

Priya: No leave, but I'll work remotely Monday/Tuesday.

Jon: Thanks, all. Let's do one last round: shout-outs and wins?

Rachel: Shout-out to Alex for fast turnaround on PDF bugfixes!

Alex: Team win! Thanks, everyone.

Priya: Shout-out to Rachel for catching those edge-cases in regression.

Jon: All right, unless anyone has AOB, let's adjourn. Thanks all—see you next week!

--- End of Meeting ---

(Breakout) Alex: Rachel, do you want to review the stack trace now or after lunch?

Rachel: I'm free now for 20 minutes.

Alex: Perfect, spinning up the debugger...

Alex: That panic in pdf_parser.go L416 is from a nil pointer on oldPage := pageBuffer[pageIndex]. Looks like it only triggers with certain scanned PDFs.

Rachel: Looking at the logs, files with 500+ pages trigger it reliably.

Alex: I'll add a print for pageIndex and file ID.

Rachel: We can write a test to fuzz parsed PDFs with random splits.

Alex: Logging a sample now.

Rachel: If it's a stale or uninitialized buffer, we need a guard clause.

Alex: Will patch and run through full regression.

Santiago: Hey folks—just dropping in from the metrics team. Did we merge the fix for chunked upload?

Bob: The PR still awaits QA approval, Santiago.

Eve: I just reviewed and approved. Should auto-merge soon. Will sync with dev at lunch.

Alice: As an aside, what's the status on the Go routine CPU bench, Tom?

Tom: I've profiled with pprof; LLM agent orchestration spiked to 105% usage. Trying to offload some calls to our Python fallback as a stopgap.

Lisa: Can we get those pprof flamegraphs posted for team review?

Tom: Sure, will do after standup.

Sara: I noticed some latency spikes on the doc-type classification endpoint. Could be due to recent traffic bursts?

Bob: Possibly—Monday was peak ingest (over 30k PDFs/hour) after marketing email went out.

Priya: We saw page load degrade when more than 10k docs were live in the queue.

Alex: I'll add per-queue metrics to the health endpoint so it's easier to monitor.

Alice: Next week's sprint will focus on semantic search and feedback integration, right?

Jon: Correct. The Product team wants more natural language queries highlighted in onboarding.

Ada: For reliability, I'll expand segment recovery tests and experiment with injected network faults.

Santiago: Quick question—anyone else seeing memory leaks on the batch extract run?

Eve: None on my jobs, but did you use the latest feature branch?

Santiago: No, I'm still on dev.

Eve: Try main, then rerun; I pushed a fix this morning.

Rachel: Just a heads up—the QA regression ran green for all submitted PRs except the CSV export bug.

Priya: I'll prioritize that fix today.

Tom: By the way, we also want to add benchmarking of the new Go routine pool next week.

Jon: Please prepare a short deck for the next retro with findings.

Bob: Can QA assign a secondary test owner for scanned table PDFs? It's a big edge case.

Rachel: Happy to. I can document and assign to Olivia.

Lisa: What's the status on doc duplication after rapid ingest?

Priya: Still seeing a few duplicates with large, scanned docs.

Alex: I suspect a race condition in the ingest queue flush.

Lisa: Let's pair and debug later today.

Santiago: Regarding metrics—should we target latency or throughput for next OKR?

Sara: Let's go with both, but favor blind throughput in the next phase so we can scale benchmarks.

Eve: On LLM integration, do we want agents to default to Go or Python first for search routing?

Alex: If Go is healthy, default there. Fallback to Python if memory hits a threshold.

Alice: Sounds good. Please document logic in the orchestration README.

Tom: Can someone reach out to Documentation to update the onboarding page?

Sara: I'll own it! Will sync after meeting.

Bob: Do we have a date for when AE5 node upgrades go live?

Jon: Infrastructure says next Wednesday, barring issues.

Priya: I'll add an item to our release checklist.

Rachel: Alex, can you run the regression suite overnight to catch the buffer bug for large multipage docs?

Alex: Absolutely.

Lisa: Another small thing—a few new joiners don't have access to the Slack #pdf-bugs channel.

Ada: I'll ping IT to sort permissions.

Alex: I found the root cause of the pageBuffer crash. Will PR a fix and tag for review.

Sara: The new TypeScript doc-type classifier demo is getting positive internal feedback.

Priya: Awesome, let's get Product input before public beta.

Tom: As for CPU, we need to monitor heat maps Friday when traffic doubles.

Jon: OK, unless anything else, let's sync again Thursday at 11am.

Bob: Works for me!

Alice: See you all then.

--- New Session: Thursday Sync ---

Jon: Welcome back, everyone! Quick roll call: today's agenda includes reviewing new dashboard metrics, PDF ingest backlog, and annotation UI bugs.

Priya: Dashboard refresh landed Tuesday. Queried median ingest time, now 12% faster this week.

Alex: We patched the annotation UI crash, but another one appears with rapidly switching document tabs.

Rachel: I updated test coverage for tab switching and multi-view concurrency. Will expand edge-case tests.

Eve: On performance, diverse doc types now all pass in the LSM ingestion pipeline; nice work, Alex.

Tom: Noticed latency spikes with multi-user queries at peak. Worth trialing connection pooling tweaks?

Jon: Good suggestion, Tom. Can you experiment with pool settings and report back Monday?

Tom: Will do!

Santiago: Metrics now isolate per-client ingest lag, so Product has clearer stratification.

Priya: On new features—a client wants to add Japanese OCR. Impact?

Sara: The underlying pipeline supports it, but LLM integration might lag until next quarter.

Eve: I ran early tests with Japanese medical PDFs. Accuracy at 92%. Need to expand training set.

Jon: Action item for Sara and Eve: gather more sample docs and collaborate on OCR tuning.

Bob: On the bulk upload page, I'm seeing some timeouts if network stalls.

Ada: I can add resumable upload and retry logic as a next sprint task.

Lisa: Got beta user feedback: annotation "comment" button is sometimes hidden on mobile.

Priya: Known issue, UI break at 320px width. Already working on it.

Rachel: Test suite flagged a new bug—negative page counts crash the Go indexer.

Alex: That one's on me, will jump on it now.

Jon: Thank you! How about product adoption? Any alpha testers responding yet?

Sara: Three of the five testers say onboarding is "much improved."

Santiago: I've logged doc ingest stats per region; Europe is growing fastest.

Priya: One last note—will demo the new filter interface at Product demo hour tomorrow.

Jon: Awesome. Any final blockers or big wins before we adjourn?

Rachel: All green in QA except for ongoing scanned table edge-case.

Bob: Appreciate the teamwork!

Jon: Same time next week, folks. Have a great rest of the day!

--- End of Extended Transcript ---